Experiment 16: Time Series Analysis of Website Traffic

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'],
index_col='Date')

decomposition = seasonal_decompose(data['Page_Views'],
model='additive')

decomposition.plot()
```



```
plt.show()

data['Page_Views_MA'] =
data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()

data['Page_Views_Z'] = (data['Page_Views'] -
data['Page_Views'].mean()) /

data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

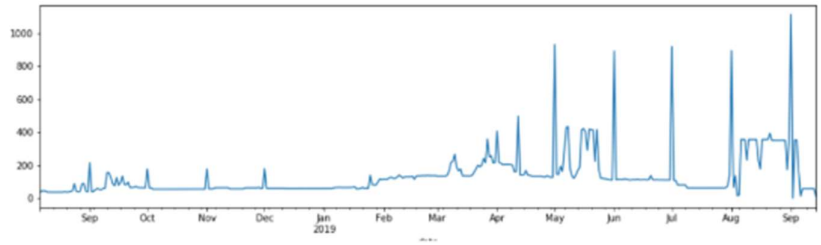
plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(anomalies.index, anomalies['Page_Views'], color='red',
label='Anomalies')
```

plt.legend()

plt.show()

OUTPUT



RESULT: Hence Time Series Analysis of Website Traffic has been successfully studied.